SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application Based Questions (20 Questions)

Course Code:	ITA0332	**Course Title:**	Mobile computing
CO Assessed:	CO1 – Imitation (BL4)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40 % of CIA	**Total Marks:**	20 (Answer ANY ONE) 1 x 20 = 20
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. (BTL 4)*		
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Section / Batch:	Slot-A/AIDS	**Date:**	28-10-2026

Code of Conduct

I CH.V.Ramayogi/192424319 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: CH.V.Ramayogi

Instructions

- This test contains 20 questions (ANSWER ANY ONE), Total: 20 Marks.
- No calculators or electronic devices permitted.
- Duration: 30 minutes.

Annexure A - QUESTIONS

1. A telecom company is deploying a cellular network in a metropolitan city with heavy user density and high-rise buildings. Analyze how frequency reuse can be optimized to maximize capacity. Evaluate the impact of co-channel interference in dense deployments. Propose a cell splitting strategy to improve network efficiency. Discuss the role of sectoring in reducing interference. Examine how user mobility affects cell planning decisions. Suggest methods to maintain signal quality in urban environments. Incorporate real-world constraints such as cost and infrastructure. Design a scalable architecture for future expansion. Justify your final cellular network design.

ANSWER:

A cellular network divides a city into small coverage areas called cells. Each cell is served by a base station, and frequencies can be reused in sufficiently separated cells. In a dense city, the design must provide high capacity while controlling interference.

Frequency reuse: Use a planned reuse pattern and assign channels so that nearby cells do not use the same frequencies unnecessarily. In modern LTE and 5G systems, frequency reuse of 1 is common, so interference is controlled mainly through power control, scheduling, beamforming and coordinated radio-resource management rather than large fixed reuse factors.

Co-channel interference: It occurs when cells using the same frequency interfere with one another. High-rise buildings cause reflections, shadowing and uneven signal levels. Interference can reduce signal-to-interference-plus-noise ratio (SINR), data rate and call quality. Interference coordination, antenna tilt, suitable transmit power and directional antennas can reduce the problem.

Cell splitting and sectoring: Split overloaded macro cells into smaller microcells or small cells in busy streets, shopping areas and transport hubs. This increases spatial reuse but adds handovers and deployment cost. Sectoring divides a cell into directional sectors, commonly three 120-degree sectors, reducing the amount of unwanted signal received from other directions.

Mobility and signal quality: Place cells along roads and transit corridors, tune handover thresholds, and use drive tests and radio-planning tools to identify coverage holes. Small cells can be installed indoors where outdoor signals are weak. Fiber or high-capacity wireless backhaul should connect base stations.

Scalable design: Begin with macro sites for broad coverage, add sectoring and targeted small cells in high-traffic areas, and expand capacity based on measured demand. Share poles and ducts where possible to reduce cost. This layered design balances coverage, capacity, interference, mobility and future growth.

2. A rural communication provider needs to establish a cellular system over a large geographic area with minimal infrastructure. Analyze the challenges of signal propagation in rural environments. Evaluate the use of macro cells versus micro cells. Propose an efficient base station placement strategy. Discuss power management techniques for long-distance coverage. Examine how terrain affects signal strength. Suggest methods to ensure connectivity with limited resources. Incorporate economic constraints in deployment decisions. Design a reliable network for sparse population density. Justify your design approach.

ANSWER:

Rural networks must cover large areas with fewer users and limited budgets. The main challenges are long distances, hills, forests, buildings and limited power or backhaul. These factors can weaken or block radio signals and create coverage gaps.

Macro cells versus microcells: Macro cells are preferable for wide rural coverage because they use elevated sites and cover larger areas. Microcells are useful only in villages, markets or other local hotspots where traffic is concentrated. Deploying many microcells across sparsely populated land would increase cost without enough users to justify it.

Base-station placement: Select elevated locations that provide line-of-sight where possible and cover roads, villages, schools, health centres and public facilities. Use propagation modelling, terrain maps and field measurements before finalizing sites. Overlap adjacent coverage areas enough to support handovers, but avoid excessive overlap that wastes resources.

Power and backhaul: Use efficient radio equipment, remote power control and sleep modes during low traffic where supported. Solar power with battery storage can serve locations with unreliable electricity. Fiber is suitable where available; microwave links or other reliable backhaul options can connect remote sites.

Terrain and coverage: Hills may require additional sites, directional antennas or carefully placed repeaters. Antenna height, frequency band and transmit power affect range. Lower-frequency bands generally travel farther and penetrate obstacles better than higher-frequency bands, although available spectrum and capacity must also be considered.

Cost-effective reliability: Share towers and backhaul among operators where regulations and commercial arrangements allow. Prioritize high-population villages and essential services first, then expand in phases using traffic and coverage data. Maintain backup power, remote monitoring and spare equipment for critical sites. A macro-cell-first design with targeted local cells offers the best balance of wide coverage, reliability and cost for sparse populations.

3. A smart city project involves connecting millions of IoT devices through a cellular network. Analyze how the network architecture must evolve to support massive connectivity. Evaluate signaling overhead in such environments. Propose efficient resource allocation techniques. Discuss the role of edge computing in reducing latency. Examine scalability challenges in IoT-based systems. Suggest improvements in protocol stack design. Incorporate reliability requirements for critical applications. Design an optimized architecture for smart city deployment. Justify your proposed solution.

ANSWER:

A smart city may connect smart meters, traffic sensors, parking systems, cameras, environmental monitors and public-safety devices. The network must support a very large number of devices, many of

which send small and infrequent messages, while critical services may require high reliability and low latency.

Architecture: Use a layered design containing IoT devices, radio access networks, a cellular core, edge-computing sites and cloud platforms. Technologies such as LTE-M and NB-IoT can support many low-power devices, while 4G/5G broadband services can support cameras and high-data-rate applications. Network slicing, where supported, can separate critical services from ordinary sensor traffic.

Signaling and resource allocation: If millions of devices attempt to connect at once, random-access procedures and control signaling can overload the network. Stagger device reporting, use efficient sleep and wake schedules, group non-urgent transmissions and apply access barring or backoff during congestion. Allocate radio resources according to traffic priority, latency needs and device capabilities.

Edge computing: Process time-sensitive data near the base station or local edge server. For example, traffic signals can respond locally to congestion without waiting for a distant cloud. The cloud can handle long-term storage, city-wide analytics and model training.

Protocol design and scalability: Use lightweight messaging such as MQTT or CoAP over suitable secure transports where appropriate. Minimize unnecessary handshakes and payload size, support device authentication and manage device firmware centrally. Monitor signaling load, packet loss, latency and battery life.

Reliability and security: Critical services need redundancy, backup power, protected communications, access control and clear recovery procedures. Separate sensitive public-safety traffic from general sensor traffic. The proposed architecture combines low-power IoT connectivity, broadband cellular service, edge processing and centralized management. It scales by adding radio capacity and edge resources as the number of devices grows, while prioritizing safety-critical applications.

4. A high-speed train experiences frequent call drops while moving across multiple cells. Analyze the handoff mechanisms involved in such scenarios. Evaluate hard and soft handoff techniques. Propose improvements to ensure seamless connectivity. Discuss the impact of speed on handoff performance. Examine network latency during transitions. Suggest methods to reduce call drops. Incorporate predictive mobility models. Design a robust handoff strategy. Justify your approach.

ANSWER:

A high-speed train crosses cell boundaries rapidly. If the next cell is not prepared before the train leaves the serving cell, a handover may fail and the call or data session may drop.

Handover mechanisms: A mobile device measures signal strength and quality from its serving cell and neighbouring cells. It reports measurements to the network, which decides when and where to hand over

the connection. A hard handover breaks the old radio link before completing the new one; it is common in LTE and 5G. Soft handover, in which a device can maintain links with multiple base stations, is associated with some CDMA systems and is not generally the standard LTE/5G handover method.

Effect of speed: At high speed, the signal changes quickly and the device may pass through a cell before handover completes. Doppler shift, rapidly changing channel conditions and signaling delay make decisions harder. Small cells placed too close together can create excessive handovers.

Improvements: Provide continuous coverage along the railway using carefully positioned trackside sites and directional antennas aimed along the track. Use a suitable cell size and overlapping coverage. Configure measurement thresholds, time-to-trigger and hysteresis carefully to prevent late handovers and ping-pong switching. Where supported, use dual connectivity or make-before-break techniques to improve continuity.

Predictive mobility: Train location, direction, speed and route timetable can help the network predict the next serving cell. The network can prepare target resources before the train reaches the boundary. Prioritize handover signaling and use low-latency backhaul between trackside sites.

Monitoring: Measure handover success rate, interruption time, dropped-call rate and radio quality along the route. Test during peak and off-peak conditions. A route-aware handover strategy with planned trackside coverage and early target preparation reduces call drops and improves passenger connectivity.

5. A telecom operator wants to enhance video streaming quality over cellular networks. Analyze key performance metrics such as latency and throughput. Evaluate the impact of network congestion on streaming. Propose QoS mechanisms to improve performance. Discuss adaptive bitrate streaming techniques. Examine protocol-level optimizations. Suggest improvements in network resource allocation. Incorporate user experience considerations. Design a solution for consistent video delivery. Justify your design.

ANSWER:

Video streaming quality depends mainly on throughput, latency, jitter, packet loss and the stability of the connection. Throughput determines the amount of video data delivered per second. High latency and jitter can delay playback and control messages, while packet loss may cause stalls or lower picture quality.

Congestion: During busy periods, many users compete for limited radio resources and backhaul capacity. The result may be lower throughput, buffering and frequent changes in video quality. Congestion can occur in the radio access network, transport network or content-delivery path.

Quality of Service (QoS): Classify traffic and manage resources according to service requirements. Schedule real-time and interactive traffic appropriately, while avoiding unfair starvation of other users. Use admission control, congestion monitoring and sufficient backhaul capacity. QoS mechanisms can improve treatment of traffic but cannot create unlimited capacity.

Adaptive bitrate streaming: The video player monitors available throughput and buffer level, then selects a suitable video bitrate. It lowers quality when bandwidth falls and raises quality when conditions improve. Shorter segments can improve adaptation speed but may add overhead. A suitable buffer helps reduce stalls without causing excessive startup delay.

Protocol and network optimization: Use efficient transport protocols, caching and content-delivery networks near users. Reduce retransmission delays where appropriate, tune congestion control and prioritize timely delivery of video segments. Allocate radio resources dynamically based on measured load and user conditions.

User experience: Track startup time, rebuffering ratio, average resolution and playback failures, not throughput alone. Test on different devices, cell-edge locations and busy-hour conditions. The recommended solution combines adaptive bitrate streaming, local content caching, congestion-aware scheduling, reliable backhaul and continuous performance monitoring. This approach improves consistency without promising the same video quality under every radio condition.

6. A mobile device frequently switches between 4G and 5G networks. Analyze how protocol stacks manage interoperability. Evaluate challenges in session continuity. Propose solutions for seamless handover. Discuss compatibility issues between technologies. Examine signaling overhead during transitions. Suggest optimization techniques for performance. Incorporate user experience considerations. Design a unified protocol framework. Justify your approach.

ANSWER:

A device that moves between 4G LTE and 5G must preserve its data sessions, authentication state and application experience while changing radio access or core-network paths. Interworking depends on the deployed architecture, device capability, operator configuration and coverage.

Protocol-stack interoperability: Both systems use layered protocols for radio access, mobility management, session handling and user data. LTE and 5G NR have different radio procedures, while 5G can operate with different core-network arrangements. The network coordinates measurements, mobility decisions, security and session management.

Continuity challenges: A device may encounter different coverage, authentication procedures, supported frequency bands, data rates and latency. If the target network is unavailable or the session is

not mapped correctly, packets can be delayed or lost. Repeated switching can also create signaling overhead and battery drain.

Solutions: Use standardized inter-system mobility and handover procedures supported by the operator network. Maintain compatible subscriber credentials and security policies. Where supported, use dual connectivity or make-before-break procedures to reduce interruption. Ensure IP-session continuity and coordinate the radio access network with core-network session management.

Reducing signaling: Configure measurement events and thresholds carefully to avoid unnecessary switching. Use accurate neighbour-cell lists and optimize coverage at 4G/5G boundaries. Devices should support required bands and software updates. Network operators should monitor handover success, interruption time, re-registration frequency and session failures.

Unified framework: A practical design includes multi-mode devices, interoperable radio access, coordinated mobility management, consistent authentication, session continuity and policy control. During transition, the network should select the best available access according to coverage, load, service needs and device capability. This design reduces interruptions while preserving security and a predictable user experience.

7. A stadium experiences heavy network congestion during events. Analyze traffic distribution challenges. Evaluate load balancing techniques across cells. Propose dynamic spectrum allocation strategies. Discuss the role of small cells in high-density areas. Examine interference management methods. Suggest improvements in network capacity. Incorporate real-time traffic analytics. Design an optimized network solution. Justify your design.

ANSWER:

A stadium may contain tens of thousands of users who simultaneously upload photos, stream clips, use messaging and access event applications. The resulting traffic surge can overload nearby cells and backhaul links.

Traffic distribution: A single macro cell may not provide enough capacity for the crowd. Users are also unevenly distributed among seating sections, entrances, concourses and hospitality areas. Network planning should estimate user density, expected traffic per user and busy-hour demand.

Small cells and sectoring: Deploy indoor or outdoor small cells and distributed antenna systems where suitable. Divide the venue into carefully planned sectors so that radio resources can be reused spatially. Directional antennas and appropriate power levels help limit interference between neighbouring sectors.

Load balancing: Use measurements of cell load, radio quality and user mobility to distribute users among available cells and carriers. Carrier aggregation and additional spectrum can increase capacity

when supported. Dynamic spectrum allocation and coordinated scheduling should respond to real-time demand while controlling interference.

Analytics and QoS: Monitor active users, throughput, PRB utilization, packet loss, latency and backhaul usage. Predict traffic peaks based on event schedules and adjust capacity before gates open. Prioritize safety, event operations and emergency communications according to policy, while managing ordinary traffic fairly.

Implementation: Perform a radio survey, install extra capacity at high-density locations, provide high-capacity fiber or equivalent backhaul and test with simulated crowds. Coordinate frequency plans and antenna placement to avoid creating interference. Use temporary cells for exceptional events if permanent capacity would be uneconomic. This layered plan—macro coverage, sectoring, small cells, load balancing and live analytics—improves capacity and service quality during major events.

8. A security breach is detected in a mobile network affecting communication protocols. Analyze vulnerabilities in protocol stacks. Evaluate authentication and encryption mechanisms. Propose enhancements for secure communication. Discuss intrusion detection techniques. Examine risks associated with signaling attacks. Suggest methods to strengthen network security. Incorporate privacy considerations. Design a secure protocol architecture. Justify your approach.

ANSWER:

A breach affecting mobile communication protocols may arise from weak authentication, implementation flaws, insecure signaling, misconfigured equipment or attacks that overload network control functions. Security must protect confidentiality, integrity, availability and subscriber privacy.

Protocol vulnerabilities: Attackers may exploit unauthenticated or poorly validated messages, replay old signaling, impersonate network elements, abuse roaming interfaces or flood signaling procedures. Legacy technologies may provide weaker protection than newer systems. Incorrect configuration and unpatched network functions increase risk.

Authentication and encryption: Use standardized mutual authentication where supported, secure key generation and storage, and strong encryption and integrity protection for signaling and user data. Keys should be derived and refreshed through approved procedures. Encryption alone is not enough: integrity checks, replay protection and correct identity validation are also important.

Detection and prevention: Apply rate limits, signaling firewalls, anomaly detection and intrusion detection systems. Monitor unusual registration bursts, repeated authentication failures, abnormal location updates and traffic spikes. Segment network functions, restrict management interfaces and apply least-privilege access. Patch software, rotate credentials and audit configuration regularly.

Signaling resilience: Use capacity limits, overload controls and filtering at network boundaries. Establish incident-response procedures to isolate affected components, preserve logs, restore trusted configurations and notify responsible teams.

Privacy: Minimize exposure of permanent subscriber identifiers, restrict access to location data and retain logs only as required by policy and law. Protect management and analytics systems as well as the radio interface.

Secure architecture: Combine secure authentication, encryption and integrity protection with network segmentation, signaling protection, continuous monitoring, updates and recovery plans. Test the design through authorized security assessments. This layered approach is more effective than relying on one security mechanism and helps maintain service during attempted attacks.

9. A manufacturer designs a mobile station supporting multiple radio technologies. Analyze hardware-software interaction in such systems. Evaluate power consumption challenges. Propose energy-efficient design techniques. Discuss antenna and RF module integration. Examine processing requirements for multi-mode support. Suggest improvements in device architecture. Incorporate performance optimization methods. Design a high-efficiency mobile station. Justify your design.

ANSWER:

A multi-mode mobile station may support 2G, 3G, 4G, 5G, Wi-Fi, Bluetooth and satellite or positioning functions. Integrating these technologies creates challenges in radio-frequency (RF) design, antenna placement, processing, heat and battery use.

Hardware-software interaction: The modem, RF transceiver, power amplifiers, filters, antennas and operating-system software must coordinate to select bands, control transmission power and manage handovers. Poor coordination can cause interference, excess heat or unnecessary radio activity.

Power consumption: Transmitting at high power, scanning multiple networks, maintaining several radio links and processing high-rate data all consume energy. Weak coverage can increase transmit power and cause repeated network searches. Multiple active radios can also create coexistence problems.

Energy-saving techniques: Use dynamic transmit-power control, efficient modem sleep states, discontinuous reception and transmission where supported, and turn off unused radios. The software should scan intelligently instead of searching continuously. Applications should batch non-urgent network requests and avoid unnecessary background traffic.

RF and antenna integration: Use antennas designed for supported frequency bands, with suitable isolation and matching. Antenna diversity and multiple-input multiple-output (MIMO) improve reception and throughput but must be managed with power and thermal limits in mind. RF front-end components should minimize insertion loss.

Processing and architecture: A power-efficient modem, optimized drivers and hardware acceleration can reduce energy per bit. Thermal monitoring should limit performance safely when the device becomes hot. Software updates and radio testing are necessary to maintain compatibility.

Evaluation: Measure battery drain, energy per transmitted bit, connection success, throughput and temperature under realistic movement and coverage conditions. The best design selects the suitable radio based on service requirements, signal quality, cost and energy. Efficient RF integration combined with intelligent software control provides reliable multi-mode operation and longer battery life.

10. A telecom provider upgrades its legacy network to high-speed data services. Analyze required architectural changes. Evaluate packet switching advantages. Propose migration strategies. Discuss backward compatibility challenges. Examine infrastructure upgrade requirements. Suggest cost-effective solutions. Incorporate performance improvement goals. Design an efficient transition plan. Justify your approach.

ANSWER:

Upgrading a legacy mobile network requires changes to radio access, core-network architecture, transport, operations and subscriber devices. Traditional circuit-switched systems reserve resources for calls, whereas packet switching sends data in packets and shares network resources more efficiently for bursty internet traffic.

Architectural changes: Deploy modern radio access equipment, packet-based core functions and higher-capacity backhaul. LTE uses an all-IP architecture for data and voice services such as VoLTE, while 5G introduces a service-based core in standalone deployments. The exact upgrade depends on the existing network and target technology.

Packet switching advantages: It uses capacity more efficiently for internet traffic, supports a wide range of IP applications and enables flexible routing and service integration. Quality of service and traffic management are still required for delay-sensitive applications.

Migration plan: First audit existing equipment, spectrum, sites, backhaul, subscriber devices and traffic. Upgrade core and transport capacity, deploy new radio equipment in stages, and test coverage, handovers, emergency calling, billing and security. Start with high-demand locations and expand based on results.

Backward compatibility: Older devices may not support new bands or services. Maintain legacy access temporarily where commercially and technically necessary, and provide clear device and SIM migration guidance. Interworking and voice continuity must be tested before legacy services are retired.

Cost control: Reuse towers, power systems, fiber routes and site facilities where possible. Prioritize software upgrades when sufficient, but replace equipment that cannot meet capacity or security requirements. Train operations staff and maintain rollback plans.

Performance evaluation: Track throughput, latency, dropped calls, coverage, availability and cost per unit of traffic. A phased migration reduces service disruption and spreads investment over time. The recommended approach modernizes the core and backhaul alongside radio access, while protecting existing subscribers until the new network is stable.

11. A telecom company is deploying a cellular network in a metropolitan city with heavy user density and high-rise buildings. Analyze how frequency reuse can be optimized to maximize capacity. Evaluate the impact of co-channel interference in dense deployments. Propose a cell splitting strategy to improve network efficiency. Discuss the role of sectoring in reducing interference. Examine how user mobility affects cell planning decisions. Suggest methods to maintain signal quality in urban environments. Incorporate real-world constraints such as cost and infrastructure. Design a scalable architecture for future expansion. Justify your final cellular network design. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

For a dense metropolitan deployment, the main objective is to increase capacity without creating excessive co-channel interference, handover failures or unaffordable site costs. The design should be based on measured traffic, building layout and expected growth.

Begin with a macro-cell layer to provide broad coverage and mobility. Use frequency reuse and radio-resource planning appropriate to the technology; LTE and 5G commonly reuse the same spectrum across cells, so interference coordination, scheduling and beamforming become especially important. Monitor SINR, throughput and cell-edge performance.

Identify overloaded cells using busy-hour traffic and user counts. Split these cells by adding microcells or small cells in hotspots such as business districts, stations and shopping streets. Avoid indiscriminate cell splitting because smaller cells increase handover frequency, site rental costs and backhaul demand. Use three-sector sites where directional coverage can improve capacity, and optimize antenna tilt and transmit power.

High-rise buildings create shadowing and multipath. Use indoor systems or targeted small cells in buildings where needed, and validate designs with propagation models and drive tests. Configure handover thresholds to support moving users without ping-pong switching. Use high-capacity fiber or reliable alternative backhaul and synchronize sites where required by the radio technology.

To manage cost, share ducts, poles and tower infrastructure when possible and prioritize sites according to capacity benefit. Design the network modularly so more radios, spectrum or small cells can be added later. Review coverage, interference, throughput, dropped-session rate and cost per added user regularly. A macro-plus-small-cell architecture, supported by sectoring, interference coordination and data-driven expansion, is scalable and economically practical.

12. A rural communication provider needs to establish a cellular system over a large geographic area with minimal infrastructure. Analyze the challenges of signal propagation in rural environments. Evaluate the use of macro cells versus micro cells. Propose an efficient base station placement strategy. Discuss power management techniques for long-distance coverage. Examine how terrain affects signal strength. Suggest methods to ensure connectivity with limited resources. Incorporate economic constraints in deployment decisions. Design a reliable network for sparse population density. Justify your design approach. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

A rural network must provide dependable coverage over a wide area while keeping capital, power and maintenance costs low. Because population density is low, the design should prioritize coverage and essential routes rather than placing many small cells.

Use macro base stations at elevated sites with clear coverage toward villages, roads, schools and health centres. Lower-frequency spectrum is generally preferable for broad coverage and better penetration, where it is available and permitted. Terrain maps and propagation modelling should guide site selection, followed by field testing to confirm actual signal levels. Hills, forests and valleys may require additional sites or directional antennas.

Microcells should be installed only at localized traffic centres, such as a busy village market or public facility. Solar panels with battery storage can support remote sites with unreliable electricity. Use efficient equipment, remote alarms and energy-saving modes during low traffic where supported. Connect sites through fiber where available or engineered microwave backhaul where fiber is impractical.

Limited budgets require phased implementation. First connect the largest communities and key transport routes, then extend service based on demand and coverage measurements. Infrastructure sharing can reduce tower, power and backhaul expenses. Keep spare parts and remote monitoring available to limit costly maintenance visits.

Reliability should be evaluated through coverage probability, call setup success, dropped calls, data availability, backup-power duration and repair time. A cost model should compare each proposed site's expected benefit with its construction and operating expense. A macro-cell-first plan, selective local cells, robust power and suitable backhaul provides the most practical balance between wide-area coverage, service quality and affordability.

13. A smart city project involves connecting millions of IoT devices through a cellular network. Analyze how the network architecture must evolve to support massive connectivity. Evaluate signaling overhead in such environments. Propose efficient resource allocation techniques. Discuss the role of edge computing in reducing latency. Examine scalability challenges in IoT-based systems. Suggest improvements in protocol stack design. Incorporate reliability requirements for critical applications. Design an optimized architecture for smart city deployment. Justify your proposed solution. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

A smart city cellular system must connect large numbers of low-data-rate sensors while also supporting high-bandwidth and safety-critical services. Its architecture should separate device connectivity, data transport, edge processing, cloud analytics and management.

Use NB-IoT or LTE-M for suitable low-power devices, and LTE/5G broadband for cameras, connected vehicles and high-rate systems. A cellular core should authenticate devices, manage sessions and route traffic to edge platforms and cloud services. Network slicing or equivalent traffic separation can be used where supported to isolate critical applications from ordinary sensor traffic.

Millions of devices can generate signaling storms when they reconnect simultaneously after a power or coverage failure. Use randomized backoff, access controls, staggered reporting schedules and efficient device sleep cycles. Send only necessary data and combine readings when the application permits. Allocate resources according to latency, reliability, traffic volume and battery constraints.

Edge servers should process urgent events close to the source. For example, traffic-control alerts can be evaluated locally, while the cloud stores historical data and performs city-wide analysis. Use lightweight

protocols such as MQTT or CoAP where appropriate, with authentication, encryption and controlled device provisioning.

Critical applications require redundancy, backup power, monitoring and recovery procedures. Segment public-safety systems from less sensitive services and maintain secure firmware-update mechanisms. Monitor registration failures, packet delivery, latency, device battery levels and edge-service availability.

The design should be scalable through modular radio capacity, expandable edge clusters and centralized device management. Pilot deployments should test congestion, failures and recovery before city-wide rollout. This combination of suitable IoT radio technologies, edge computing, efficient signaling and layered security supports growth while protecting critical services.

14. A high-speed train experiences frequent call drops while moving across multiple cells. Analyze the handoff mechanisms involved in such scenarios. Evaluate hard and soft handoff techniques. Propose improvements to ensure seamless connectivity. Discuss the impact of speed on handoff performance. Examine network latency during transitions. Suggest methods to reduce call drops. Incorporate predictive mobility models. Design a robust handoff strategy. Justify your approach. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

A train moving at high speed may cross several cell boundaries within minutes. The network must prepare the next cell early enough to complete handover before the current radio link becomes unreliable.

The mobile station measures serving and neighbouring cells and reports results. The network selects a target cell and coordinates radio resources and security context. Hard handover, used in LTE/5G mobility, releases the old connection as the new connection is established; soft handover is associated with some CDMA systems and should not be assumed available in LTE/5G. The key metrics are handover success rate, interruption time and call-drop rate.

At high speed, rapid signal changes, Doppler effects and signaling delays can make measurements outdated. Small cells placed too close together may cause excessive handovers. Plan trackside sites to provide continuous corridor coverage, use directional antennas and ensure sufficient overlap. Configure hysteresis and time-to-trigger to avoid both late handovers and ping-pong switching.

A predictive strategy can use train location, speed, direction and known route geometry to identify the next cell. The network can prepare the target cell and allocate resources before the train reaches the

boundary. Fast backhaul and coordinated mobility management reduce preparation delays. Dual connectivity or other continuity features may be used where supported.

Test the whole route under different speeds, weather and passenger loads. Record radio quality, handover preparation time, packet loss and service interruption. Tune cell boundaries and neighbour lists based on test results. Route-aware prediction combined with planned trackside coverage is more reliable than relying only on reactive signal thresholds.

15. A telecom operator wants to enhance video streaming quality over cellular networks. Analyze key performance metrics such as latency and throughput. Evaluate the impact of network congestion on streaming. Propose QoS mechanisms to improve performance. Discuss adaptive bitrate streaming techniques. Examine protocol-level optimizations. Suggest improvements in network resource allocation. Incorporate user experience considerations. Design a solution for consistent video delivery. Justify your design. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

Consistent mobile video requires enough throughput for the selected resolution, low enough latency for responsive playback, controlled jitter and low packet loss. The design must also respond gracefully when many users share the same cell.

Measure radio utilization, backhaul load, throughput distribution, startup delay, rebuffering ratio and average delivered resolution. Congestion can cause lower bitrates, stalls and repeated quality changes. Capacity should be evaluated at busy hour and at cell edges, not only under ideal conditions.

Adaptive bitrate streaming is essential. The client estimates throughput and buffer occupancy, then requests a suitable video representation. When capacity falls, it selects a lower bitrate to prevent buffering; when capacity recovers, it increases quality cautiously. Buffer size and segment duration should balance smooth playback against startup delay and responsiveness.

The operator can apply QoS and fair scheduling to manage competing traffic, use admission and congestion controls, and upgrade overloaded radio or transport links. Caching popular content at a nearby CDN reduces repeated long-distance transfers and can improve delivery time. Efficient transport and congestion-control settings can help, but cannot overcome a fundamental lack of radio capacity.

Use real-time analytics to identify congested cells and dynamically allocate available resources. Prioritize essential services according to policy without unfairly starving other users. Test across device types, mobility conditions and different video bitrates. A combined solution of adaptive streaming, local

caching, fair resource allocation, sufficient backhaul and experience-based monitoring provides better video quality than optimizing throughput alone.

16. A mobile device frequently switches between 4G and 5G networks. Analyze how protocol stacks manage interoperability. Evaluate challenges in session continuity. Propose solutions for seamless handover. Discuss compatibility issues between technologies. Examine signaling overhead during transitions. Suggest optimization techniques for performance. Incorporate user experience considerations. Design a unified protocol framework. Justify your approach. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

A device that moves between LTE and 5G NR must keep voice calls, data sessions and application connections usable during radio and core-network transitions. The exact procedures depend on whether the 5G deployment uses a non-standalone or standalone architecture and on operator support.

The protocol stack handles radio measurements, mobility control, security, session management and user-plane packet delivery. LTE and 5G have different radio protocols and may use different core functions, so interworking must be planned rather than assumed. Devices also need compatible bands, software and subscriber provisioning.

Continuity problems can occur when the target network has weak coverage, lacks resources, applies different policies or cannot maintain the existing session. Transition signaling can increase latency and power use. Repeated switching near a coverage boundary can create instability.

Use standardized inter-system handover and mobility procedures, correct neighbour configuration and consistent authentication and security policies. Where supported, dual connectivity or make-before-break methods can reduce interruption. Coordinate radio access and core-network session management so the device can retain appropriate IP connectivity and application state.

Reduce unnecessary signaling by tuning measurement thresholds, hysteresis and time-to-trigger. Plan 4G/5G coverage overlap and ensure sufficient backhaul. Test voice continuity, data sessions, emergency calls, packet loss and handover delay under realistic movement. Monitor failed transitions and repeated registration attempts.

A unified operational framework should combine multi-mode devices, interoperable mobility management, consistent subscriber identity, secure session handling and centralized performance monitoring. This approach makes transitions more predictable and improves user experience while recognizing that continuity depends on the deployed network and device capabilities.

17. A telecom operator wants to enhance video streaming quality over cellular networks. Analyze key performance metrics such as latency and throughput. Evaluate the impact of network congestion on streaming. Propose QoS mechanisms to improve performance. Discuss adaptive bitrate streaming techniques. Examine protocol-level optimizations. Suggest improvements in network resource allocation. Incorporate user experience considerations. Design a solution for consistent video delivery. Justify your design. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

For consistent video delivery, the network should optimize the user's viewing experience rather than simply maximize peak throughput. The most important metrics include sustained throughput, latency, jitter, packet loss, startup time, rebuffering and delivered resolution.

Start by identifying where congestion occurs: radio access, backhaul, core network or content delivery. Use cell-level telemetry and application experience metrics to locate the bottleneck. High utilization in a single stadium or urban cell may require localized capacity, while widespread transport congestion may require a backhaul upgrade.

Use adaptive bitrate streaming so the client selects a representation that matches current network conditions. A conservative bitrate estimate and adequate playback buffer can reduce stalls. Shorter segments may improve responsiveness, but add request and protocol overhead. Cache popular content at edge or CDN locations to reduce repeated transfers across the core.

Apply fair scheduling and QoS policies so video receives appropriate treatment without starving voice, emergency or other important traffic. Dynamically allocate radio resources based on load and channel quality. Optimize transport configuration and avoid unnecessary retransmissions where application and protocol behavior allow. QoS cannot substitute for sufficient capacity, so add spectrum, carriers, small cells or backhaul where measurements justify it.

Evaluate the solution with realistic user tests during peak demand. Track rebuffering ratio, startup delay, resolution stability and failed sessions by location and device. The recommended approach combines adaptive clients, local caching, congestion-aware resource management and evidence-based capacity expansion. This provides stable playback while controlling infrastructure cost.

18. A security breach is detected in a mobile network affecting communication protocols. Analyze vulnerabilities in protocol stacks. Evaluate authentication and encryption mechanisms. Propose enhancements for secure communication. Discuss intrusion detection techniques. Examine risks associated with signaling attacks. Suggest methods to strengthen network security. Incorporate

privacy considerations. Design a secure protocol architecture. Justify your approach. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

After a mobile-network breach, the operator should first contain the incident, determine the affected protocol or component, preserve evidence and restore service from trusted configurations. Long-term remediation must address both the exploited weakness and the wider security architecture.

Possible weaknesses include insufficient message validation, weak or misconfigured authentication, replay opportunities, exposed management interfaces, signaling floods and unpatched network functions. Legacy interconnects and roaming interfaces can also create risks if they are not adequately filtered.

Use standardized mutual authentication, secure key management, encryption and integrity protection appropriate to the network generation. Protect signaling against replay and malformed messages. Apply signaling firewalls, rate limits and overload controls to reduce denial-of-service risks. Segment network functions and restrict management access using least privilege and multifactor authentication where applicable.

Deploy monitoring that detects abnormal registration patterns, repeated authentication failures, unexpected traffic routes and unusual signaling volume. Correlate logs across radio, core and management systems. Patch vulnerable software, audit configuration, rotate affected credentials and conduct authorized penetration testing after remediation.

Privacy protections should limit exposure of subscriber identities and location data, restrict access to logs and enforce retention rules. Incident-response plans should define escalation, isolation, recovery, communications and post-incident review. Redundant components and tested backups help restore service without reintroducing compromised settings.

The secure architecture is layered: trusted authentication, protected signaling and user traffic, network segmentation, continuous detection, timely updates and tested recovery. Security testing should confirm that the original attack path is closed and that defensive controls do not prevent legitimate emergency or roaming services.

19. A manufacturer designs a mobile station supporting multiple radio technologies. Analyze hardware-software interaction in such systems. Evaluate power consumption challenges. Propose energy-efficient design techniques. Discuss antenna and RF module integration. Examine processing requirements for multi-mode support. Suggest improvements in device architecture. Incorporate performance optimization methods. Design a high-efficiency mobile station. Justify

your design. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

A multi-radio device must support several standards and frequency bands while controlling battery use, heat, interference and hardware cost. Its design depends on the intended bands, simultaneous-radio requirements, antenna space and expected data rates.

The modem and operating system coordinate radio selection, measurements, handovers and power states. The RF front end contains amplifiers, filters, switches and transceivers; poor integration can increase signal loss and power consumption. Antenna placement must consider hand effects, enclosure materials, isolation and coexistence between radios.

High power use can result from weak coverage, frequent network scanning, repeated handovers, high-rate transmission and simultaneous active radios. Use dynamic transmit-power control, modem sleep modes, discontinuous reception/transmission where supported, and intelligent scanning. Applications should batch background traffic and avoid waking the radio unnecessarily.

Support for MIMO and antenna diversity can improve performance, but requires careful RF layout and thermal management. Use low-loss components, suitable antenna matching and effective shielding. Hardware acceleration and efficient modm firmware can reduce energy per bit. The operating system should select a radio based on signal quality, service needs, cost and power consumption.

Validate the device under real-world conditions: weak signal, cell-edge operation, movement, simultaneous Wi-Fi and cellular use, and sustained data transfer. Measure battery drain, throughput, call quality, temperature and handover success. Include software updates and compliance testing across supported bands.

The recommended design combines efficient RF components, well-isolated antennas, optimized modem software and power-aware radio selection. It offers reliable multi-standard connectivity while extending battery life and maintaining safe thermal performance.

20. A telecom provider upgrades its legacy network to high-speed data services. Analyze required architectural changes. Evaluate packet switching advantages. Propose migration strategies. Discuss backward compatibility challenges. Examine infrastructure upgrade requirements. Suggest cost-effective solutions. Incorporate performance improvement goals. Design an efficient transition plan. Justify your approach. (Extended scenario variation with additional constraints and optimization requirements.)

ANSWER:

A legacy network upgrade should improve data capacity and user experience without creating unnecessary service disruption or abandoning subscribers before they can migrate. The plan must address radio access, core functions, backhaul, spectrum, devices, operations and cost.

Packet switching is well suited to internet traffic because it shares network capacity among data flows and supports IP applications. Modern LTE networks use an all-IP data architecture, and 5G standalone networks use a service-based core. Voice and legacy service interworking must be planned according to the technologies being deployed.

First audit network assets, traffic patterns, spectrum, site power, backhaul capacity, subscriber devices and service obligations. Upgrade overloaded transport links and core functions, then deploy modern radio equipment in phases. Start with high-demand areas and validate coverage, throughput, latency, voice service, billing, security and emergency calling before expanding.

Backward compatibility is important during transition. Older devices may lack new frequency bands or radio features. Provide device and SIM guidance, maintain legacy coverage for an appropriate period and test handovers between old and new systems. Define clear criteria for retiring older technologies and communicate changes to customers.

Control cost by reusing towers, power systems, ducts and fiber where practical. Share infrastructure when permitted, use software upgrades where sufficient and replace obsolete or insecure components when necessary. Train staff and keep rollback plans for major changes.

Track throughput, latency, dropped calls, service availability, customer complaints and cost per unit of traffic. A staged migration with clear acceptance tests, capacity planning and subscriber support provides a safer, more economical path to high-speed services than a sudden network replacement.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____